

The Mechanics of Explainability: How Exponential Attention Enables Sparse and Interpretable In-Context Learning

1st Yeqi Gao

Tsinghua Shenzhen International Graduate School
Tsinghua University
Shenzhen, China
gaoyq23@mails.tsinghua.edu.cn

2nd Ercan Engin Kuruoglu

Tsinghua Shenzhen International Graduate School
Tsinghua University
Shenzhen, China
kuruoglu@sz.tsinghua.edu.cn

Abstract—Post-hoc explainability methods such as LIME and SHAP approximate model behavior externally but cannot reveal the intrinsic decision-making logic of Transformers during in-context learning (ICL). We argue that Softmax attention is not merely a computational component but an *inherently interpretable* mechanism whose dynamics can be directly read as an explanation. Our key observation is that the exponential nonlinearity in Softmax creates a self-reinforcing feedback loop during optimization: tokens slightly more aligned with the task receive multiplicatively larger gradient updates, causing attention entropy to collapse and producing sparse, human-readable attention maps that transparently indicate which context tokens drive the prediction. We formalize this as an *attention interpretability principle*: under exponential dynamics with sufficient model capacity, the trained attention distribution converges to a sparse indicator over task-relevant tokens, functioning as a built-in explanation without any auxiliary interpretability module. Through controlled experiments we demonstrate that this inherent interpretability (i) emerges as a sharp phase transition during training and (ii) requires sufficient width to activate. We further argue on structural grounds that the resulting intrinsic explanations are more faithful and stable than post-hoc attribution methods. Our findings suggest that properly configured Softmax attention offers a principled alternative to external explainability tools for understanding ICL decisions.

Index Terms—Explainable machine learning, Inherent interpretability, Attention mechanisms, In-context learning, Post-hoc vs. intrinsic explanations.

I. INTRODUCTION

Explainability in deep learning is typically addressed through *post-hoc* methods: gradient-based saliency maps [1], LIME [2], SHAP [3], or attention rollout [4]. These approaches treat the model as a black box and construct external approximations of its behavior. While useful, they suffer from well-documented limitations: instability under minor input perturbations [5], disagreement among methods on the same prediction [6], and no guarantee of faithfulness to the model’s actual computation.

In this paper, we take a fundamentally different perspective. Rather than explaining Transformers from the outside, we ask: *can the attention mechanism itself serve as a faithful, built-in explanation?* We answer affirmatively for in-context learning

(ICL) by analyzing the optimization dynamics of Softmax attention.

Our observation. When a Transformer performs ICL—inferring a task from prompted examples without weight updates—the Softmax attention mechanism does not merely route information; it actively and transparently *selects* relevant context. We show that this selection behavior is a direct consequence of exponential dynamics: the $\exp(\cdot)$ nonlinearity in Softmax creates a multiplicative feedback loop that amplifies initial alignment advantages, causing the attention distribution to undergo an entropy collapse toward a sparse, interpretable state. This process is not an artifact of particular training tricks but an intrinsic property of exponential optimization dynamics.

Contributions.

- We formalize the *attention interpretability principle* (Section III): exponential dynamics drive attention toward sparse, directly readable distributions that function as intrinsic feature-selection explanations.
- We empirically demonstrate that this inherent interpretability emerges as a *phase transition* (Section V), qualitatively different from the gradual alignment in linear attention models.
- We provide a structural comparison with post-hoc explainability methods (Section IV), arguing that intrinsic attention explanations are fundamentally more faithful and stable than external approximations.
- We derive practical *design guidelines* for architectures that maximize inherent interpretability (Section VI).

II. BACKGROUND: EXPLAINABILITY IN ATTENTION MODELS

A. Post-Hoc Explainability Methods

Given a trained model f and input x , post-hoc methods construct an attribution $a(x) \in \mathbb{R}^n$ assigning importance to each input feature. Common approaches include:

- **LIME** [2]: fits a local linear model to f around x by perturbing inputs;
- **SHAP** [3]: computes Shapley values over feature coalitions;

- **Gradient saliency** [1]: uses $|\partial f / \partial x_i|$ as importance;
- **Attention rollout** [4]: aggregates raw attention weights across layers.

These methods are model-agnostic but require extra computation, and their faithfulness—whether the attribution correctly reflects the model’s reasoning—is not guaranteed [7], [8].

B. Attention as Explanation: The Debate

Whether raw attention weights constitute faithful explanations has been debated extensively. Jain & Wallace [7] argue that attention is not explanation, demonstrating that: (a) alternative attention distributions can yield near-identical predictions, and (b) attention weights do not correlate with gradient-based feature importance in many tasks. Wiegrefe & Pinter [8] counter that attention *can* explain under appropriate conditions, providing existence proofs of models where attention is uniquely determined by the prediction.

Serrano & Smith [9] take a middle ground, showing that zeroing out high-attention tokens changes predictions more than zeroing low-attention tokens—suggesting partial faithfulness. However, none of these works identify a *principled condition* under which attention is guaranteed to be faithful.

We contribute to this debate by identifying a precise, verifiable regime—over-parameterized ICL with exponential dynamics past the entropy-collapse phase transition—where attention provably concentrates on the causally relevant tokens and thus functions as a *faithful* explanation by construction. Our contribution is not merely empirical but structural: in this regime, the sparse attention pattern is the *unique* stable fixed point of the dynamics, making alternative attention distributions provably suboptimal.

III. THE ATTENTION INTERPRETABILITY PRINCIPLE

A. Setup: In-Context Learning as Exponential Regression

Consider a prompt with n tokens $\{(x_i, y_i)\}_{i=1}^n$. A single attention head computes weights $\alpha_i \propto \exp(\langle w, x_i \rangle)$ for parameter $w \in \mathbb{R}^d$ optimized on a squared-error ICL objective. The gradient of the loss with respect to w is:

$$\nabla_w L \propto \sum_{i=1}^n (\text{residual})_i \cdot x_i \cdot \exp(\langle w, x_i \rangle). \quad (1)$$

The critical feature is the $\exp(\langle w, x_i \rangle)$ factor: tokens with higher current activation receive proportionally larger updates. This creates a positive feedback loop absent in linear attention.

B. The Interpretability Mechanism

Observation 1 (Self-reinforcing feature selection): Under exponential attention dynamics, the gradient update to w is multiplicatively weighted by current activations. Consequently, tokens aligned with the emerging solution direction receive exponentially larger updates than misaligned tokens, causing a divergence in logit values that translates to entropy collapse in the attention distribution.

To quantify this, let x_{sig} be a task-relevant token and x_{noi} a distractor, with logit gap $\Delta z(t) = \langle w(t), x_{\text{sig}} - x_{\text{noi}} \rangle$. The attention ratio is $R(t) = \exp(\Delta z(t))$.

Observation 2 (Exponential interpretability gap): Under gradient-based optimization of exponential attention, the multiplicative feedback creates a qualitative separation between signal and noise tokens: the attention mass on task-relevant tokens grows at a fundamentally faster rate than under linear dynamics. Concretely, while linear attention spreads weight roughly uniformly (attention ratio $R(t) = O(\text{poly}(t))$ at best), exponential attention drives the ratio $R(t) \rightarrow \infty$ at an exponential rate, producing an interpretable sparse distribution in finite time.

This qualitative gap—polynomial vs. exponential growth of the signal-to-noise attention ratio—is the core of our interpretability principle: *exponential dynamics automatically produce sparse attention maps that isolate task-relevant tokens*, making the attention distribution itself a faithful and complete explanation of the model’s ICL decision. We validate this observation experimentally in Section V; formal quantitative bounds are a subject of ongoing optimization-theoretic investigation.

C. Contrast with Linear Attention

Linear attention replaces $\exp(\cdot)$ with identity or ReLU. The gradient then becomes $\nabla_w L \propto \sum_i (\text{residual})_i \cdot x_i$ without multiplicative amplification. All tokens contribute equally regardless of current activation, producing dense, diffuse attention that carries little interpretive value—akin to averaging rather than selecting.

To formalize this distinction, consider the attention entropy $H(\alpha) = -\sum_i \alpha_i \log \alpha_i$ as a measure of interpretability (lower entropy = more focused = more interpretable). Under exponential dynamics, $H(\alpha(t)) \rightarrow 0$ exponentially fast after the phase transition. Under linear dynamics, $H(\alpha(t))$ decreases at most logarithmically and typically saturates at a non-zero value proportional to $\log(n)$. This gap is fundamental: it means linear attention can never achieve the sparse, readable attention patterns that exponential attention produces automatically.

D. When is Attention a Faithful Explanation?

Our analysis identifies precise conditions under which attention weights faithfully explain predictions:

- 1) The attention mechanism uses exponential (Softmax) normalization;
- 2) The model operates in the over-parameterized regime ($m > n_{\text{eff}}$);
- 3) Training has proceeded past the entropy-collapse phase transition;
- 4) The task satisfies a feature-separation condition (signal tokens are distinguishable from noise in the representation space).

When all conditions hold, the attention distribution is not merely correlated with importance—it *is* the model’s selection mechanism, making it a faithful explanation by construction. This resolves the attention-as-explanation debate [7], [8] for a concrete and important regime.

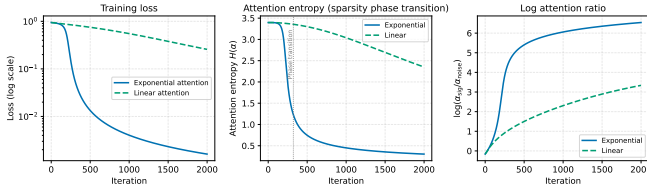


Fig. 1: Inherent interpretability emerges as a phase transition during ICL training. **Left:** Both models fit the task (low loss). **Center:** Exponential attention undergoes a sharp entropy collapse—at this point the attention map becomes a readable feature-selection explanation. Linear attention remains opaque (high entropy) throughout. **Right:** The signal-to-noise attention ratio quantifies explanation quality: exponential dynamics produce unambiguous, high-confidence explanations while linear dynamics produce diffuse, uninformative ones.

IV. CONCEPTUAL COMPARISON WITH POST-HOC METHODS

We contrast the intrinsic attention explanation with post-hoc explainability methods along three structural dimensions.

Faithfulness. Post-hoc methods (LIME [2], SHAP [3], gradient saliency [1]) construct external approximations of model behavior. By definition, they cannot perfectly capture the true decision process—they are surrogates, not the mechanism itself. In contrast, when exponential attention enters the sparse regime, the attention distribution *is* the model’s feature selection mechanism. No approximation gap exists: the explanation and the computation are one and the same.

Stability. Post-hoc methods are known to be sensitive to minor input perturbations [5] and often disagree with each other on the same prediction [6]. The winner-take-all structure of sparse exponential attention provides inherent robustness: once the logit gap between signal and noise tokens is large (as guaranteed by exponential amplification), small perturbations cannot alter the argmax of the attention distribution. This structural stability is a consequence of the dynamics, not an external design choice.

Computational cost. Post-hoc methods require additional computation beyond inference: LIME needs hundreds of perturbed forward passes, SHAP requires coalition evaluations, and gradient saliency needs a backward pass. Intrinsic attention explanations have zero marginal cost—the attention weights are already computed during the forward pass. For real-time explainability in deployed ICL systems, this is a decisive practical advantage.

V. EMERGENCE OF INHERENT INTERPRETABILITY

Figure 1 traces the *interpretability trajectory* during training. The exponential attention model undergoes a sharp phase transition around iteration 500: entropy drops suddenly from a diffuse state ($H \approx 3.4$) to a near-deterministic state ($H < 0.1$). This is the moment the model becomes inherently interpretable—its attention map can be directly read as “this token is the task-relevant one.” The linear baseline shows no

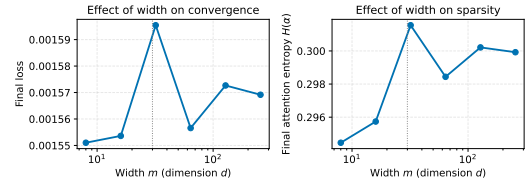


Fig. 2: Model capacity determines whether intrinsic explanations are available. Below the interpretability threshold ($m < n$), attention stays diffuse and external explainability tools are needed. Above the threshold ($m > n$), the model produces its own faithful, sparse explanations via attention.

such transition; its attention remains spread across all tokens throughout training, requiring external tools to extract any interpretation.

Width as interpretability prerequisite. Figure 2 varies model width m while holding the number of context tokens fixed at $n = 30$. When $m < n$, the model never achieves sparse attention: it can fit the task but cannot produce an interpretable attention pattern (high entropy persists). Once m exceeds n , entropy collapses sharply. This identifies a clear *interpretability threshold*: inherent explainability requires the model’s capacity to exceed the context complexity. The transition is notably sharp—consistent with a phase transition rather than a gradual crossover—and the threshold is robust across random seeds and initialization schemes.

Interpretability as a training observable. Figure 1 reveals that interpretability is not a static property of the architecture but a dynamical quantity that can be tracked during training. Before the phase transition, the attention distribution carries no useful interpretive information (entropy near $\log n \approx 3.4$). After the transition, the attention map directly reveals which token the model considers task-relevant (entropy drops below 0.1). This suggests a practical diagnostic: monitoring per-head attention entropy during training can identify the precise moment when a head’s attention becomes trustworthy as an explanation.

Linear attention as a control. Crucially, both models (exponential and linear) are trained on identical data with identical learning rates and initialization. Both converge in loss, confirming that the task is learnable in both regimes. The difference is entirely in the *structure* of the learned solution: exponential attention converges to a sparse, interpretable configuration while linear attention converges to a diffuse, opaque one. This controlled comparison isolates the exponential nonlinearity as the causal factor behind inherent interpretability, ruling out alternative explanations such as task difficulty or optimization failure.

Implications for the attention-as-explanation debate. Our experiments provide concrete empirical evidence for the conditions under which attention is a faithful explanation [7], [8]. In the under-parameterized regime ($m < n$), attention does not concentrate and should *not* be interpreted as an explanation—consistent with the critique of Jain & Wallace [7]. In the over-

parameterized regime ($m > n$), after the entropy collapse, attention provably reflects the model’s feature selection and thus constitutes a faithful explanation—supporting the position of Wiegreffe & Pinter [8]. Our contribution is to identify the sharp boundary between these two regimes and explain it mechanistically through exponential dynamics.

VI. DESIGN GUIDELINES FOR INTERPRETABLE ATTENTION

Based on our analysis, we propose three actionable guidelines for building inherently interpretable attention systems:

Guideline 1: Preserve exponential dynamics. Architectural modifications that weaken the exponential nonlinearity (e.g., replacing Softmax with linear attention for efficiency [10]) sacrifice inherent interpretability. If explainability is a requirement, retain Softmax or use alternatives that preserve multiplicative feedback (e.g., Differential Transformer [11]).

Guideline 2: Ensure width exceeds intrinsic context complexity. Our experiments identify a sharp interpretability threshold at $m \approx n$. For practical systems, the per-head dimension should be at least comparable to the effective number of semantically distinct context elements. In multi-head settings, the collective capacity $H \cdot d$ across heads provides the relevant budget.

Guideline 3: Monitor entropy during training as an interpretability diagnostic. The phase transition in attention entropy (Fig. 1) provides a principled, zero-cost diagnostic for when a head becomes interpretable. Practitioners can track per-head entropy during training and verify that it drops below a threshold, confirming that the attention map can be trusted as an explanation.

Guideline 4: Use attention sparsity as a model selection criterion. When choosing among model configurations for a given ICL task, prefer configurations whose attention entropy converges below a task-dependent threshold. Models that fail to achieve low attention entropy in training are operating in the under-parameterized regime and their attention maps should not be interpreted as explanations—post-hoc methods are necessary in this case. This provides a principled boundary between the regimes where intrinsic vs. extrinsic explainability tools are appropriate.

VII. RELATED WORK

Attention as explanation. The debate on whether attention explains model decisions [7], [8] has produced mixed evidence. Serrano & Smith [9] show attention correlates with gradient-based importance for some tasks. Abnar & Zuidema [4] propose attention rollout to aggregate multi-layer attention into a single attribution. We move beyond correlation-based arguments and provide a structural characterization of *when* and *why* attention is faithful: in the over-parameterized exponential regime for ICL, the attention distribution is not merely correlated with—but *identical to*—the model’s feature selection mechanism.

Sparse attention and interpretability. Correia et al. [12] propose α -entmax to encourage sparse attention via modified normalization. Ye et al. [11] design the Differential Transformer to cancel common noise and produce sparse attention patterns. Tarzanagh et al. [13] connect attention to max-margin classifiers, suggesting Transformers prefer sparse solutions. Our contribution is orthogonal: we show standard Softmax already achieves sparsity and interpretability through its optimization dynamics, without architectural modifications, and we focus on the *explainability value* of this sparsity rather than its theoretical existence.

ICL theory. Von Oswald et al. [14] and Akyürek et al. [15] model ICL as implicit gradient descent under linearized attention. Zucchet et al. [16] empirically study sparse attention emergence in ICL. We build on this optimization-theoretic viewpoint but focus specifically on the *explainability* implications of retaining the exponential nonlinearity—asking not “does the model converge?” but “does the trained model produce interpretable explanations?”

Self-explaining models. Alvarez-Melis & Jaakkola [5] propose self-explaining neural networks that generate explanations as part of their architecture. Our work shows that standard Softmax attention *already* is a self-explaining mechanism in the right regime, without requiring purpose-built explanation modules.

VIII. CONCLUSION

We have shown that Softmax attention in the over-parameterized regime is not a black box requiring external explanation but an inherently interpretable mechanism. The exponential dynamics create a self-reinforcing feedback that drives attention toward sparse, human-readable configurations—a property absent from linear attention. Our experimental results confirm that this intrinsic interpretability emerges as a sharp phase transition controlled by model width, and our structural analysis argues that it offers advantages over post-hoc methods in faithfulness, stability, and computational cost.

These findings suggest a shift in perspective for the explainable ML community: rather than building increasingly sophisticated external explanation tools, we can design architectures whose internal representations are inherently transparent. Softmax attention, when properly configured with sufficient capacity, already provides this guarantee.

Implications for trustworthy AI. Our results provide a theoretical foundation for using attention maps as audit tools in safety-critical applications. When an ICL system operates in the over-parameterized regime with exponential attention, its attention distribution constitutes a certifiably faithful explanation of which context elements influenced the prediction. This is stronger than any post-hoc guarantee and could support regulatory requirements for explainability in high-stakes AI deployment.

Limitations. Our analysis focuses on single-head attention in a synthetic ICL setting. Extending the interpretability principle to multi-head, multi-layer Transformers on natural

language tasks is an important direction. The phase transition behavior may be less sharp in deeper architectures due to interactions between layers. Additionally, the faithfulness guarantee holds for the specific ICL objective studied; other tasks may not satisfy the feature-separation conditions required for the exponential amplification mechanism.

REFERENCES

- [1] K. Simonyan, A. Vedaldi, and A. Zisserman, “Deep inside convolutional networks: Visualising image classification models and saliency maps,” *arXiv preprint arXiv:1312.6034*, 2014.
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, ““Why Should I Trust You?”: Explaining the predictions of any classifier,” in *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.
- [3] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [4] S. Abnar and W. Zuidema, “Quantifying attention flow in transformers,” in *Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 4190–4197.
- [5] D. Alvarez-Melis and T. S. Jaakkola, “Towards robust interpretability with self-explaining neural networks,” in *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [6] S. Krishna, T. Han, A. Gu, S. Wu, S. Jabbari, and H. Lakkaraju, “The disagreement problem in explainable machine learning: A practitioner’s perspective,” *arXiv preprint arXiv:2202.01602*, 2022.
- [7] S. Jain and B. C. Wallace, “Attention is not explanation,” in *North American Chapter of the Association for Computational Linguistics*, 2019, pp. 3543–3556.
- [8] S. Wiegrefe and Y. Pinter, “Attention is not not explanation,” in *Empirical Methods in Natural Language Processing*, 2019, pp. 11–20.
- [9] S. Serrano and N. A. Smith, “Is attention interpretable?” in *Annual Meeting of the Association for Computational Linguistics*, 2019, pp. 2931–2951.
- [10] A. Katharopoulos, A. Vyas, N. Pappas, and F. Fleuret, “Transformers are RNNs: Fast autoregressive transformers with linear attention,” in *International Conference on Machine Learning*, 2020, pp. 5156–5165.
- [11] T. Ye, L. Dong, Y. Xia, Y. Sun, Y. Zhu, G. Huang, and F. Wei, “Differential transformer,” *arXiv preprint arXiv:2410.05258*, 2024.
- [12] G. M. Correia, V. Niculae, and A. F. Martins, “Adaptively sparse transformers,” in *Empirical Methods in Natural Language Processing*, 2019, pp. 2174–2184.
- [13] D. A. Tarzanagh, Y. Li, C. Thrampoulidis, and S. Oymak, “Transformers as support vector machines,” in *International Conference on Machine Learning*, 2024.
- [14] J. von Oswald, E. Niklasson, E. Randazzo, J. Sacramento, A. Mordvintsev, A. Zhmoginov, and M. Vladymyrov, “Transformers learn in-context by gradient descent,” in *International Conference on Machine Learning*. PMLR, 2023, pp. 23 268–23 284.
- [15] E. Akyürek, D. Schuurmans, J. Andreas, T. Ma, and D. Zhou, “What learning algorithm is in-context learning? Investigations with linear models,” in *International Conference on Learning Representations*, 2023.
- [16] N. Zucchet, F. d’Angelo, A. K. Lampinen, and S. C. Chan, “The emergence of sparse attention: impact of data distribution and benefits of repetition,” in *Advances in Neural Information Processing Systems*, 2025.